

MXCuBE site report

P11, DESY (Hamburg, Germany)

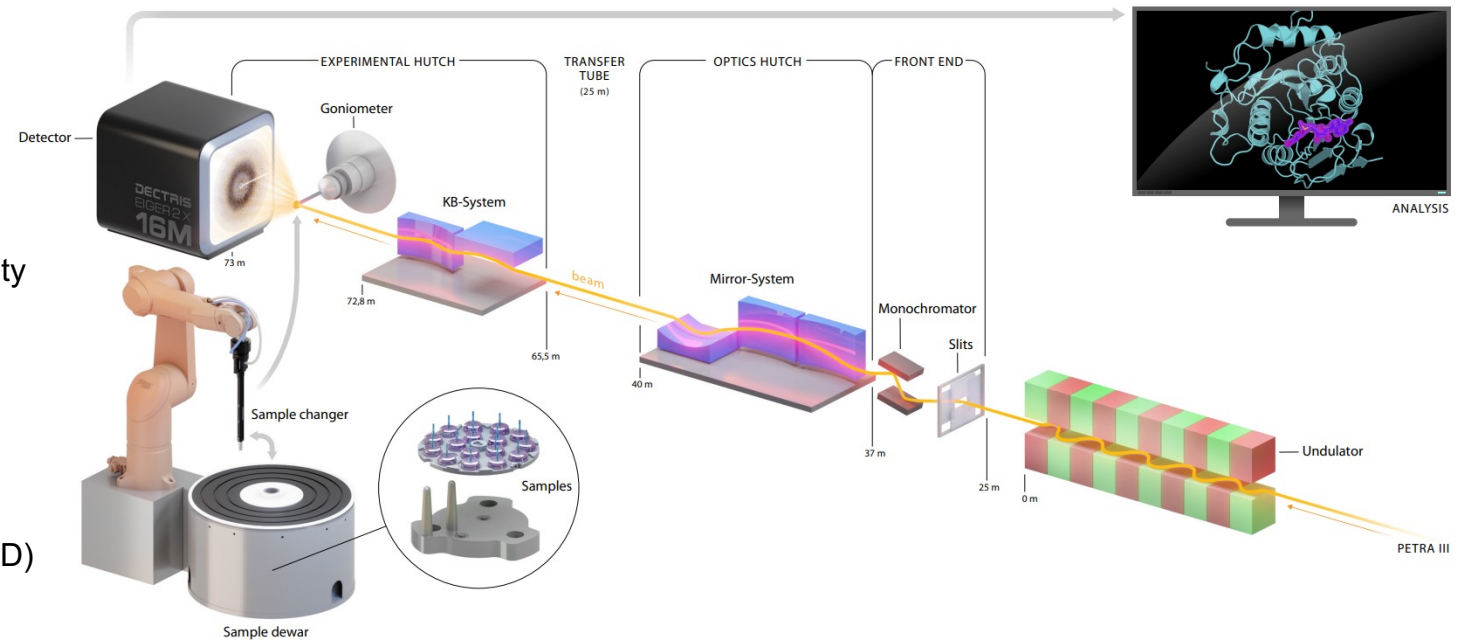
Brief beamline summary

The High-throughput Macromolecular Crystallography Beamline P11



Crystallography Experiments

- In user operation since 2013
- Broad energy range: 5.5 - 28 keV
- High-speed sample changer with capacity (23 unipucks = 368 samples)
- High precision single axis goniometer: 0.0001° at $120^\circ/\text{s}$
- Eiger2 X 16M
- XRF for experimental phasing (SAD/MAD) and element analysis



MXCuBE

- Latest mxcubeqt (dev) with a bunch of local changes – PR pending
- ~~Latest (1.273.0)~~ mxcubecore **1.193.0** (develop branch)
- Python 3.11.9, PyQt 5.15.10.

Developments since last meeting

- MXCuBE is now the default GUI of the beamline!
- Operation testing and hot fixes
- Qt and Core versions are frozen
- New LIMS component and xml2yaml pending until further notice
- Some PRs are still pending – thanks for patience!

Plans for the next six months

- ✓ Roll-out to the users
- ✓ Interaction with ISPyB
- ✓ EDNA Characterization + processing – ~~needs testing after cluster update~~
 - X-Ray centering + mesh-scans
 - Hot fixes
 - New PRs
 - Optimize the usage of *murko*-centering
 - ~~Code cleanup~~
 - ~~Move installation to Debian 12~~
 - ~~Aligning the temporary *murko* setup with the local IT infrastructure~~